User stories completed this sprint:

SCRUM-30: As a user, I want to earn badges by inputting data, so that I can be rewarded for my contributions

SCRUM-48: As a user, I want to be able to keep track of certain attributes, like the number of uploads and my uploads streak, so I can 'gamify' my experience

SCRUM-49: As a user, I want to be able to view a leaderboard with other users' streaks and upload count, so I can be encouraged to contribute to the app

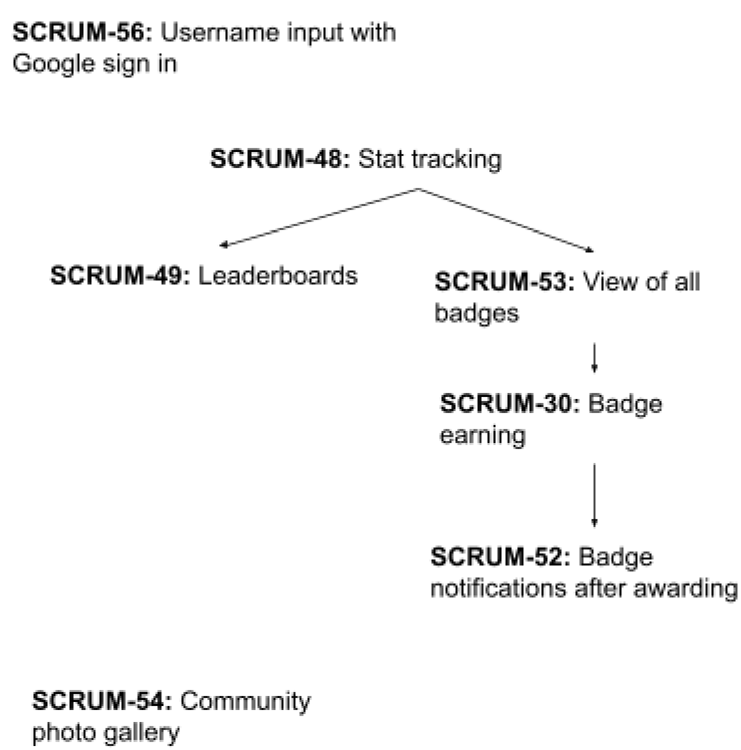
SCRUM-53: As a user, I want to see a list of all available badges with descriptions, so I know what actions I can take to earn them.

SCRUM-52: As a user, I want to receive a notification when I earn a new badge, so that I feel rewarded for my actions.

SCRUM-56: As a user, I want to enter a username when logging in with google, so that I can keep track of my account

SCRUM-54: As a user, I want to view a photo gallery of community uploads sorted by popularity or recency, so I can explore interesting locations.

Network Diagram



Dependencies

- **SCRUM-48:** Track user stats
  - Enables the following:
    - **SCRUM-53:** View of all badges
      - Only after creating all the badges would it be possible to implement the succeeding user stories
    - **SCRUM-49**

Critical Path

SCRUM-48 -> **SCRUM-53 -> SCRUM-30 -> SCRUM-52**

The badge notification system followed a tightly coupled critical path that began with implementing core stat tracking in SCRUM-48. This provided the backend infrastructure needed to monitor user actions like upload count and streaks. SCRUM-53 then introduced a view of all available badges, serving as an incentive for defining badge conditions. With that in place, SCRUM-30 handled the logic for awarding badges based on user activity. Finally, SCRUM-52 completed the path by delivering notifications when a badge was earned. Each step depended on the previous one, and delays in tracking or logic would have blocked downstream progress. This made the path essential for delivering a cohesive and rewarding badge system during the sprint.

Risk Areas (things that could've gone wrong)

Several components of the stat tracking and badge system carried potential risks that could have disrupted the sprint if not properly managed.

- **SCRUM-48** introduced backend stat tracking, which was foundational for all badge-related functionality. Any inaccuracies in tracking user activity could have led to incorrect badge awards or gaps in leaderboard data.
- **SCRUM-30,** which handled the logic for earning badges, depended on real-time and reliable access to those stats. This posed a risk of logic errors or misfires if data was incomplete or delayed.

To mitigate these risks, we ensured early coordination between backend and frontend tasks, clarified data flows across components, and established test cases for key scenarios. As a result, the interconnected nature of these stories was managed effectively, and no significant issues came up during development.